

Dark Matter Module: the physics behind

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1 Dark Matter

- **Target topics:** Dark Matter (galaxy rotation curves, clusters).
- **Learning objectives:** Analyze multiple lines of evidence for dark matter; derive galaxy rotational velocity curve from mechanics, Newton gravitation law and Kepler law; investigate galaxies and clusters composition; find the cluster mass from virial theorem reviewing kinetic energy and gravitational potential; connect classical Physics to Cosmology.
- **Physics program link:** Mechanics, Kepler's law, Newton's law of gravitation, centripetal acceleration, circular motion, centripetal force, gravitational force, energy and momentum conservation, gravitational potential energy, kinetic energy.

1.1 Galaxy rotational velocity curve

- **Objectives:** Apply the law of gravitation to derive the expected orbital velocity in a galaxy. Compare the theoretical velocity with observational data from galaxy datasets. Introduce the dark matter hypothesis as a potential solution to the discrepancy.
- **Dataset variables of galaxies:** [17]
 - Rad (radius)
 - V_{obs} (observed velocity)
 - $\text{err}V$ (velocity error)
 - V_{gas} (gas velocity)
 - V_{disk} (stellar disk velocity)

- V_{bul} (bulge velocity)
- SB_{disk} (surface brightness disk)
- SB_{bul} (surface brightness bulge)

Classical Physics derivation:

- We model a small mass m (orbiting body) in a stable circular orbit around a large central mass M (e.g., galactic center) at a radius r .
- The condition for stable orbit requires the Gravitational Force (F_G) to equal the Centripetal Force (F_c).

Gravitational and Centripetal Forces

- **Gravitational Force (F_G):**

$$F_G = \frac{GMm}{r^2} \quad (1)$$

- **Centripetal Force (F_c):**

$$F_c = \frac{mv^2}{r} \quad (2)$$

- **Equating the Forces ($F_G = F_c$):**

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad (3)$$

Deriving the rotational velocity (v)

- **Cancellation of m :** The mass of the orbiting body (m) cancels out, showing v is independent of m .

$$\frac{GM}{r^2} = \frac{v^2}{r} \quad (4)$$

- **Isolating v^2 :** Multiply by r to isolate v^2 .

$$v^2 = \frac{GM}{r} \quad (5)$$

- **Final Rotational Velocity (v):** Take the square root.

$$v = \sqrt{\frac{GM}{r}} \quad (6)$$

- **Physical Meaning:** This Keplerian velocity profile dictates that $v \propto 1/\sqrt{r}$, meaning orbital speed decreases as distance from the central mass increases.

Connection to Kepler's third law

- **Orbital Period (T):** Period is the circumference divided by velocity.

$$T = \frac{2\pi r}{v} \quad (7)$$

- **Kepler's relation :** Square T and substitute $v^2 = GM/r$.

$$T^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^2}{GM/r} \implies T^2 = \frac{4\pi^2}{GM} r^3 \quad (8)$$

- **Physical Meaning:** This confirms Kepler's Third Law ($T^2 \propto r^3$), demonstrating that the proportionality derived from the force balance is fully consistent with the relationship between orbital period and radius.

Velocity and mass

- **Baryonic velocity (from data):** Incorporating the Mass-to-Light ratio (Υ) and conserving the signs for counter-rotating components [16]:

$$V_{\text{bar}}^2(r) = V_{\text{gas}}|V_{\text{gas}}| + \Upsilon [V_{\text{disk}}|V_{\text{disk}}| + V_{\text{bul}}|V_{\text{bul}}|] \quad (9)$$

- **Baryonic mass:**

$$M_{\text{bar}}(r) = \frac{r \max(V_{\text{bar}}^2(r), 0)}{G} \quad (10)$$

- **Total observed mass:**

$$M_{\text{obs}}(r) = \frac{r V_{\text{obs}}^2(r)}{G} \quad (11)$$

- **Dark matter velocity squared:** Derived from the difference between the observed total velocity and the baryonic prediction:

$$V_{\text{DM}}^2(r) = V_{\text{obs}}^2(r) - V_{\text{bar}}^2(r) \quad (12)$$

- **Dark matter density (from data):**

$$\rho_{\text{DM}}(r) = \frac{1}{4\pi G r^2} \frac{d}{dr} (r \cdot V_{\text{DM}}^2(r)) \quad (13)$$

NFW Dark Matter Profile

- **NFW profile for dark matter [20, 19]:**

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \quad (14)$$

where r_s is the scale radius and ρ_s is the characteristic density, $\alpha = 1, \beta = 3, \gamma = 1$.

- **Global Fit parameters:** The structural parameters ρ_s , r_s , and the mass-to-light ratio Υ are calculated via a global χ^2 minimization on the entire rotation curve dataset to provide a robust estimation of the halo.

- **NFW dark matter mass:**

$$M_{\text{NFW}}(r) = 4\pi \int \rho_{\text{NFW}} r^2 dr = 4\pi \rho_s r_s^3 \left[\ln \left(1 + \frac{r}{r_s} \right) - \frac{r/r_s}{1 + r/r_s} \right] \quad (15)$$

- **Velocity dark matter:**

$$V_{\text{DM}}(r) = \sqrt{\frac{GM_{\text{NFW}}(r)}{r}} \quad (16)$$

- **Total simulated curve:**

$$V_{\text{sim}}(r, f) = \sqrt{\frac{GM_{\text{bar}}(r) + f \cdot M_{\text{NFW}}(r)}{r}} \quad \text{with } f \in [0, 2] \quad (17)$$

Galaxy	$M_{\text{tot}} [M_{\odot}]$	$\%M_{\text{DM}}/M_{\text{tot}}$
NGC0024	3.18×10^{10}	72.6%
NGC0055	2.56×10^{10}	85.9%
NGC0100	1.97×10^{10}	85.4%
NGC0247	3.83×10^{10}	89.2%
NGC0289	4.48×10^{11}	79.5%
NGC0300	2.59×10^{10}	86.4%
NGC0801	6.36×10^{11}	56.2%
NGC0891	1.85×10^{11}	71.4%
NGC1003	9.24×10^{10}	85.6%
NGC1090	1.84×10^{11}	70.7%
NGC1705	7.28×10^9	85.3%
NGC2366	4.09×10^9	76.3%
NGC2403	8.66×10^{10}	85.9%
NGC2683	1.86×10^{11}	74.8%
NGC3198	2.26×10^{11}	80.8%

Table 1: Total mass and DM fraction for each galaxy.

Galaxy - NGC3198

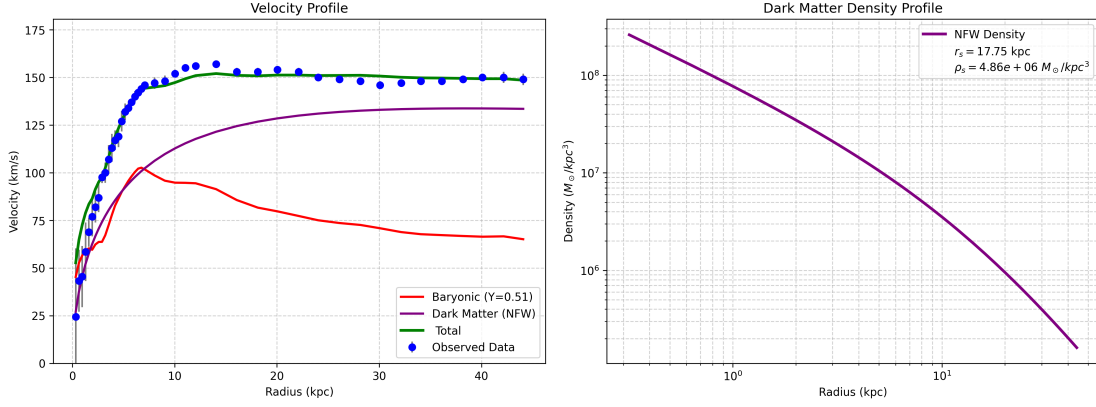


Figure 1: Galaxy velocity rotation curve and density.

Data visualization and fit

- **Plot:**

- **X-axis:** Rad (radius, data)
- **Y-axis:**
 - * V_{bar} (red)
 - * V_{obs} (data, blue) $\pm \text{errV}$
 - * V_{sim} (green)

- **Fit χ^2/dof (chi-squared):**

$$\chi^2/\text{dof} = \frac{1}{N - P} \sum_{i=1}^N \left(\frac{V_{\text{obs}}(r_i) - V_{\text{sim}}(r_i)}{\text{errV}(r_i)} \right)^2 \quad (18)$$

where N is the number of data points and P is the number of free parameters ($P = 3$: ρ_s, r_s, Υ).

Global χ^2 Minimization

The parameters r_s , ρ_s , and the mass-to-light ratio Υ are determined simultaneously by minimizing the χ^2 over the entire rotation curve:

$$\chi^2 = \sum_{i=1}^N \left(\frac{V_{\text{obs}}(r_i) - V_{\text{tot,sim}}(r_i)}{\text{errV}(r_i)} \right)^2 \quad (19)$$

where $V_{\text{tot,sim}}(r) = \sqrt{\max(V_{\text{bar}}^2(r) + V_{\text{DM}}^2(r), 0)}$.

1.2 Velocity distribution in galaxy clusters

- **Objectives:** Help students infer the existence of dark matter in galaxy clusters using classical mechanics and virial theorem. Students will apply the virial theorem to estimate the total mass of a galaxy cluster from the kinetic and potential energy of its galaxies.
- **Dataset Coma cluster (SDSS) :** objid (galaxy ID), ra (right ascension, α), dec (declination, δ), modelmag (apparent magnitude in r band, m_r), modelmagerr, extinction, redshift (z), zErr (redshift error) [22, 1].
- **Dataset Abell Clusters:** Cluster (name), ID (galaxy ID), RAdeg (right ascension in degrees), DEdeg (declination in degrees), RV (radial velocity in km/s), e_RV (velocity error in km/s), q_RV (quality flag), bmag (apparent magnitude in B band). [25]

1. **Individual Galaxy Velocity:** Assuming non-relativistic speeds for low-redshift clusters, the radial velocity of each galaxy is derived directly from its observed redshift data:

$$v_i = c z_i \quad (20)$$

2. **Number of observed galaxies:** N

3. **Mean Cluster Velocity:** The systemic velocity of the cluster is calculated as the average of the individual radial velocities:

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i \quad (21)$$

4. **Observed Velocity Dispersion:** The 1-D velocity dispersion is the standard deviation of the galaxies' velocities relative to the systemic mean, providing a measure of the cluster's dynamical state:

$$\sigma_{\text{obs}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})^2} \quad (22)$$

5. **Comoving Distance:** To determine physical scales, we integrate over the expansion history of the Universe (Eq. 15 [14]):

$$\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + (1-\Omega_m)}} \quad (23)$$

6. **Angular Diameter Distance:** This converts the comoving distance into a metric that relates apparent angular size to true physical size (Eq. 18 [14]):

$$D_A(z) = \frac{\chi(z)}{1+z} \quad (24)$$

7. **Cluster Center Identification:** The dynamical center is approximated using either the coordinates of the Brightest Cluster Galaxy (BCG) or the spatial median of all cluster members

[3, 7]:

$$(\alpha_c, \delta_c) = (\alpha_{\text{BCG}}, \delta_{\text{BCG}}) \quad \text{or} \quad (\text{median}(\alpha), \text{median}(\delta))$$

8. **Angular Separation (θ):** Applying the spherical law of cosines, we compute the angular distance between the i -th galaxy and the cluster center [8]. Note that right ascension (α) and declination (δ) must be converted to radians:

$$\theta_i = \arccos(\sin \delta_c \sin \delta_i + \cos \delta_c \cos \delta_i \cos(\alpha_c - \alpha_i)) \quad (25)$$

9. **Projected Physical Radius:** The angular separation is converted into a transverse physical distance using the angular diameter distance:

$$r_{\text{proj},i} = \theta_i D_{A,\text{kpc}} \quad (26)$$

10. **Critical Density:** The density required for a spatially flat Universe:

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G} \quad (27)$$

11. **Virial Mass Estimation (M_{200}):** Assuming the cluster is a relaxed, dynamically thermalized system, the virial theorem links the observed velocity dispersion to the total mass enclosed within the characteristic radius R_{200} [7]:

$$\sigma_{\text{obs}}^2 = \frac{GM_{200}}{3R_{200}}$$

$$M_{200} = \frac{4}{3}\pi 200 \rho_{\text{crit}} R_{200}^3 \quad (28)$$

Dark Matter Mass

22. **Method 1: Dark Matter mass subtraction.** Because the dynamical mass (M_{200}) derived from the Virial Theorem is significantly larger than the luminous mass, the Dark Matter content can be found simply by subtracting the baryonic mass:

$$M_{\text{DM}} = M_{200} - M_{\text{lum}} \quad (29)$$

To dynamically simulate the cluster using an interactive mass fraction slider (f):

$$M_{\text{tot,sim}} = M_{\text{lum}} + f M_{\text{DM}} \quad (30)$$

23. **Method 2: Dark Matter mass via NFW profile.** Alternatively, the dark matter distribution can be modeled analytically using a Navarro-Frenk-White (NFW) profile:

- *Halo Concentration Parameter* derived from empirical scaling relations (Eq. 4 [11]):

$$c = A \left(\frac{M_{200}}{M_{\text{pivot}}} \right)^B (1+z)^C \quad (31)$$

where $A = 5.71$, $B = -0.084$, and $C = -0.47$.

- *NFW Density Characteristic Factor* (Eq. 2 [20]):

$$\delta_c(c) = \frac{200}{3} \frac{c^3}{\ln(1+c) - c/(1+c)} \quad (32)$$

- *Characteristic Radius* (r_{200}) and *Scale Radius* (r_s):

$$r_{200} = \left(\frac{3M_{200}}{4\pi \cdot 200 \rho_{\text{crit}}} \right)^{1/3}, \quad r_s = \frac{r_{200}}{c} \quad (33)$$

- *Characteristic Density* (ρ_s):

$$\rho_s = \delta_c(c) \rho_{\text{crit}}$$

- *Enclosed Dark Matter Mass (NFW)*: The total mass of dark matter contained within the projected radius $r_{\text{proj},i}$:

$$M_{\text{NFW}}(r_{\text{proj},i}) = M_{200} \frac{\ln(1+x_i) - x_i/(1+x_i)}{\ln(1+c) - c/(1+c)}, \quad x_i = \frac{r_{\text{proj},i}}{r_s} \quad (34)$$

To dynamically simulate the cluster using the interactive mass fraction slider (f) with the NFW profile:

$$M_{\text{tot,sim}} = M_{\text{lum}} + f M_{\text{NFW}}(r_{\text{proj},i}) \quad (35)$$

24. **Total Simulated Velocity Dispersion:** Depending on the chosen method, the theoretical velocity dispersion is derived from the total simulated mass ($M_{\text{tot,sim}}$):

$$\sigma_{\text{sim}} = \sqrt{\frac{GM_{\text{tot,sim}}}{3R_{200}}} \quad (36)$$

25. **Observed Histogram:**

$$H_{\text{obs}} = \text{hist}(v_{\text{obs}})$$

26. **Generate Simulated Velocities:** Extract a normal probability distribution representing the galaxy kinematics, centered on the mean cluster velocity ($\bar{v}$):

$$v_{\text{sim}} \sim \mathcal{N}(\mu = \bar{v}, \sigma = \sigma_{\text{sim}})$$

27. **Simulated Histogram:**

$$H_{\text{sim}} = \text{hist}(v_{\text{sim}})$$

Cluster	$M_{\text{tot}} [M_{\odot}]$	$\%M_{\text{DM}}/\text{tot}_1$	$\%M_{\text{DM}}/\text{tot}_2$
Abell0080	1.70×10^{14}	92.5%	88.1%
Abell0118	1.10×10^{15}	91.1%	91.0%
Abell0380	5.47×10^{17}	68.2%	59.1%
Abell0487	2.83×10^{17}	72.3%	67.4%
Abell2734	1.82×10^{15}	90.1%	90.6%
Abell2778	5.63×10^{16}	80.2%	81.1%
Abell2799	2.36×10^{14}	93.1%	90.2%
Abell2800	2.29×10^{14}	92.5%	88.8%
Abell2819	1.10×10^{14}	93.6%	88.6%
Abell2854	2.38×10^{14}	92.6%	89.2%
Abell2871	1.98×10^{16}	84.1%	86.3%
Abell2911	2.51×10^{17}	73.0%	68.9%
Abell3864	1.03×10^{17}	77.6%	77.4%
Abell3921	1.00×10^{15}	91.2%	90.9%
Abell4067	3.82×10^{17}	70.5%	63.8%
coma_data	2.84×10^{15}	89.1%	90.4%

Table 2: Total mass and DM fraction (Method 1: Virial subtraction, Method 2: NFW profile) for each cluster.

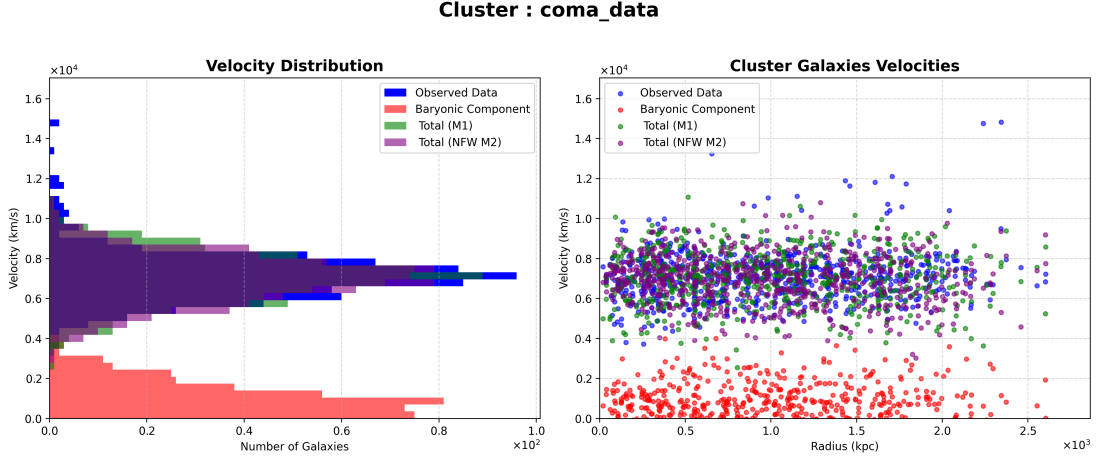


Figure 2: Coma cluster velocity.

Luminous Mass

17. **Luminosity Distance** ($z \ll 1$): For local clusters, we can approximate the luminosity distance using the linear Hubble law evaluated at the median redshift of the cluster [8]:

$$D_L \approx \frac{c z_{\text{cluster}}}{H_0}, \quad z_{\text{cluster}} = \text{median}(z_i) \quad (37)$$

18. **Distance Modulus (μ):** The difference between apparent and absolute magnitude, where D_L is expressed in parsecs (Eq. 25 [14]):

$$\mu = 5 \log_{10}(D_L) - 5 \quad (38)$$

19. **Absolute Magnitude (M_r):** Derived from the apparent magnitude in the r-band (m_r), corrected for the distance modulus and the line-of-sight Galactic extinction (A_r) [8]:

$$M_r = m_r - A_r - \left[5 \log_{10} \left(\frac{D_L}{10 \text{ pc}} \right) \right] \quad (39)$$

20. **Galaxy Luminosity (L_r):** Converted from absolute magnitude and expressed in units of Solar Luminosity ($L_\odot$):

$$L_r = 10^{0.4(M_{r\odot} - M_r)} \quad (40)$$

21. **Stellar Mass per Galaxy ($M_{\text{star},i}$):** The stellar mass is estimated by applying a theoretical Mass-to-Light ratio ($\Upsilon_r = M/L \approx 2$) typical for early-type galaxies in clusters [4]:

$$M_{\text{star},i} = \Upsilon_r L_r \quad (41)$$

22. **Total Stellar Mass:** The sum of the individual stellar masses of all observed cluster members:

$$M_{\text{star}} = \sum_{i=1}^N M_{\text{star},i} \quad (42)$$

23. **Intracluster Gas Mass:** The X-ray emitting hot gas constitutes a major fraction of the cluster's baryons. This is estimated using empirical scaling relations from observations (Section 3 [13]) 1.2. The original paper defines this relation in terms of M_{500} . However, by applying the conversion factor $M_{500} \approx 0.7 M_{200}$ 1.2 (assuming $h = 0.7$), we can modify the formula to express the gas mass directly as a function of M_{200} :

$$M_{\text{gas}} = (0.7 M_{200}) \cdot 0.093 \left(\frac{0.7 M_{200}}{2 \cdot 10^{14} M_\odot} \right)^{0.21} \quad (43)$$

24. **Total Baryonic (Luminous) Mass:** The sum of the stellar component and the intracluster medium (gas):

$$M_{\text{lum}} = M_{\text{star}} + M_{\text{gas}} \quad (44)$$

25. **Characteristic Baryonic Radius (r_{bar}):** Defined analogously to r_{200} :

$$r_{\text{bar}} = \left(\frac{3 M_{\text{lum}}}{4\pi \cdot 200 \rho_{\text{crit}}} \right)^{1/3} \quad (45)$$

26. **Baryonic Velocity Dispersion:** The theoretical velocity dispersion predicted if the cluster's

dynamics were governed solely by visible baryonic matter (no dark matter):

$$\sigma_{\text{bar}} = \sqrt{\frac{GM_{\text{lum}}}{3r_{\text{bar}}}} \quad (46)$$

Velocity Dispersion and Simulation

25. **Total simulated mass:**

$$M_{\text{tot}} = M_{\text{lum}} + f M_{200} \quad (47)$$

26. **Total simulated velocity dispersion:**

$$\sigma_{\text{sim}} = \sqrt{\frac{GM_{\text{tot}}}{3r_{200}}} \quad (48)$$

27. **Observed histogram:**

$$H_{\text{obs}} = \text{hist}(v_{\text{obs}})$$

28. **Generate normal distribution of simulated velocities:**

$$\sigma_{\text{tot}} = \text{Normal}(\mu = \sigma_{\text{sim}}, \sigma = \sigma_{\text{obs}}, N)$$

29. **Simulated histogram:**

$$H_{\text{sim}} = \text{hist}(\sigma_{\text{tot}})$$

Data visualization

- Plot histograms: number of galaxies on Y-axis, velocity on X-axis.
- Curves: observations (blue), simulated (green), baryonic (red).

Cluster Mass via Virial Theorem

1. **Angular Separation** ($\Delta\theta$) from right ascension (α) and declination (δ) data (converted to radians, where c denotes the cluster center and i the individual galaxy):

$$\Delta\theta = \arccos[\sin(\delta_c)\sin(\delta_i) + \cos(\delta_c)\cos(\delta_i)\cos(\alpha_c - \alpha_i)] \quad (49)$$

2. **Projected Radius** (R_P) from angular separation, assuming a distance $D = 100$ Mpc (radius converted to kpc):

$$R_P = D \cdot \Delta\theta \quad (50)$$

3. **Observed Velocity** (V_{obs}) from redshift data (or radial velocity RV):

$$V_{\text{obs}} = c \cdot z \quad \text{or} \quad V_{\text{obs}} = \text{RV} \quad (51)$$

4. **1-D Velocity Dispersion** (σ_{1D}) (observation):

$$\sigma_{1D} = \sqrt{\frac{1}{N} \sum_{i=1}^N (V_i - \bar{V})^2} \quad (52)$$

5. **3-D Velocity Dispersion** (σ_{3D}) (assuming isotropy):

$$\sigma_{3D} = 3 \sigma_{1D} \quad (53)$$

Baryonic components

6. **Convert Distance D (in Mpc) to pc** and compute the **Distance Modulus** (μ):

$$\mu = 5 \log_{10} \left(\frac{D_{\text{pc}}}{10 \text{ pc}} \right) \quad (54)$$

7. **Absolute Magnitude** (M) using apparent magnitude ($m = \text{bmag}$):

$$M = m - \mu \quad (55)$$

8. **Luminosity** (L) of each galaxy from absolute magnitude (M):

$$L = 10^{-0.4(M - M_{\odot})} \quad (56)$$

9. **Total Luminosity of the cluster** (L_{total}):

$$L_{\text{total}} = \sum_{i=1}^N L_i \quad (57)$$

10. **Stellar Mass** (M_{lum}) from luminosity (Υ is the mass-to-light ratio):

$$M_{\text{lum}} = \Upsilon \cdot L_{\text{total}} \quad (58)$$

11. **Gas mass:** By converting the original M_{500} relation into M_{200} (where $M_{500} \approx 0.7 M_{200}$ and $h = 0.7$) 1.2 1.2, the formula becomes:

$$M_{\text{gas}} = (0.7 M_{200}) \cdot 0.093 \left(\frac{0.7 M_{200}}{2 \cdot 10^{14} M_{\odot}} \right)^{0.21} \quad (59)$$

12. **Total Baryonic Mass:**

$$M_{\text{lum}} = M_{\text{star}} + M_{\text{gas}} \quad (60)$$

Notes:

The formula used to estimate the intracluster gas mass derived from Equation 12 in Giodini et al. (2009) [13].

Step 1: Definition of the gas mass fraction

By definition, the gas mass fraction within the radius R_{500} is the ratio of the gas mass to the total mass enclosed within that same radius:

$$f_{500}^{\text{gas}} = \frac{M_{\text{gas}}}{M_{500}} \quad (61)$$

From this definition, the gas mass can be isolated as:

$$M_{\text{gas}} = M_{500} \cdot f_{500}^{\text{gas}} \quad (62)$$

Step 2: Giodini's empirical relation (Eq. 12)[13]

An empirical scaling relation to calculate the gas fraction f_{500}^{gas} as a function of the total mass M_{500} :

$$f_{500}^{\text{gas}} \left(\frac{h}{0.7} \right)^{3/2} = (9.3 \times 10^{-2}) \left(\frac{M_{500}}{2 \times 10^{14} M_{\odot}} \right)^{0.21} \quad (63)$$

Step 3: Applying the Hubble parameter ($h = 0.7$)

The dimensionless Hubble parameter is set to $h = 0.7$. Substituting this value into the left side of Giodini's equation simplifies the scaling factor:

$$\left(\frac{0.7}{0.7} \right)^{3/2} = 1^{3/2} = 1 \quad (64)$$

Consequently, the equation for the gas mass fraction reduces to:

$$f_{500}^{\text{gas}} = 0.093 \left(\frac{M_{500}}{2 \times 10^{14} M_{\odot}} \right)^{0.21} \quad (65)$$

Step 4: Gas mass and M_{200} conversion

Substituting the simplified gas fraction expression (Eq. 65) back into the initial mass definition (Eq. 62) yields the original applied formula from the paper:

$$M_{\text{gas}} = M_{500} \cdot 0.093 \left(\frac{M_{500}}{2 \cdot 10^{14} M_{\odot}} \right)^{0.21} \quad (66)$$

To express this relation directly in terms of M_{200} , we substitute $M_{500} \approx 0.7M_{200}$ (as demonstrated in Note 1.2) into the equation:

$$M_{\text{gas}} = (0.7M_{200}) \cdot 0.093 \left(\frac{0.7M_{200}}{2 \cdot 10^{14} M_{\odot}} \right)^{0.21} \quad (67)$$

Notes:

The mathematical derivation to obtain the conversion factor between M_{500} and M_{200} relies on the analytical properties of the Navarro-Frenk-White (NFW) density profile.

Step 1: The NFW density profile and enclosed mass

The NFW density profile is defined as:

$$\rho(r) = \frac{\rho_s}{\left(1 + \frac{r}{r_s}\right)^2} \quad (68)$$

where ρ_s is the characteristic density and r_s is the scale radius. Integrating this density profile over a spherical volume up to a radius R gives the enclosed mass $M(R)$:

$$M(R) = 4\pi\rho_s r_s^3 \left[\ln\left(1 + \frac{R}{r_s}\right) - \frac{R/r_s}{1 + R/r_s} \right] \quad (69)$$

By defining the concentration parameter as $c = R/r_s$, the mass function is usually written as $M(c) \propto f(c)$, where:

$$f(c) = \ln(1 + c) - \frac{c}{1 + c} \quad (70)$$

Step 2: Definition of mass by overdensity

In cosmological models, the mass of a halo is defined as the mass enclosed within a radius R_Δ where the average internal density is Δ times the critical density of the Universe (ρ_{crit}). The mass is given by:

$$M_\Delta = \frac{4\pi}{3} \Delta \rho_{\text{crit}} R_\Delta^3 \quad (71)$$

Applying this definition to $\Delta = 500$ and $\Delta = 200$, we can express the ratio of the two masses geometrically:

$$\frac{M_{500}}{M_{200}} = \frac{500}{200} \left(\frac{R_{500}}{R_{200}} \right)^3 = 2.5 \left(\frac{c_{500}}{c_{200}} \right)^3 \quad (72)$$

where $c_{500} = R_{500}/r_s$ and $c_{200} = R_{200}/r_s$.

Step 3: Equating the NFW Mass and the overdensity mass

The mass ratio must also satisfy the NFW integrated mass equation from Step 1:

$$\frac{M_{500}}{M_{200}} = \frac{f(c_{500})}{f(c_{200})} = \frac{\ln(1 + c_{500}) - \frac{c_{500}}{1 + c_{500}}}{\ln(1 + c_{200}) - \frac{c_{200}}{1 + c_{200}}} \quad (73)$$

Equating the geometric ratio (Step 2) to the NFW mass ratio yields the conversion equation:

$$2.5 \left(\frac{c_{500}}{c_{200}} \right)^3 = \frac{\ln(1 + c_{500}) - \frac{c_{500}}{1 + c_{500}}}{\ln(1 + c_{200}) - \frac{c_{200}}{1 + c_{200}}} \quad (74)$$

Step 4: Solving for a fixed concentration ($c_{200} = 5$)

Assuming a concentration parameter $c_{200} = 5$, we evaluate the denominator of the right side:

$$f(5) = \ln(1 + 5) - \frac{5}{1 + 5} = \ln(6) - \frac{5}{6} \approx 1.7917 - 0.8333 = 0.9584 \quad (75)$$

The equation becomes:

$$2.5 \left(\frac{c_{500}}{5} \right)^3 = \frac{\ln(1 + c_{500}) - \frac{c_{500}}{1 + c_{500}}}{0.9584} \quad (76)$$

Solving this equation numerically for the unknown c_{500} yields:

$$c_{500} \approx 3.305 \quad (77)$$

Step 5: Mass ratio calculation

Substitute $c_{500} \approx 3.305$ back into the geometric mass ratio equation:

$$\frac{M_{500}}{M_{200}} = 2.5 \left(\frac{3.305}{5} \right)^3 = 2.5 \times (0.661)^3 \approx 2.5 \times 0.2888 \approx 0.722 \quad (78)$$

Therefore, for a halo with a concentration $c = 5$, the mass M_{500} is approximately **72.2%** of M_{200} , which justifies the standard approximation $M_{500} \approx 0.7 \cdot M_{200}$. **References in the**

Literature:

- **White (2001) [26]:** This paper provides the analytical background for comparing mass definitions and explicitly plots the ratio M_i/M_{200} for $c = 5$. In Figure 1, the curve corresponding to M_{500} is a horizontal line constant at ≈ 0.72 independent of Ω_m .
- **Hu & Kravtsov (2003) [15]:** Appendix C ("Mass Conversion") of this paper details the analytical transformation between different overdensity mass definitions assuming an NFW profile. Equations (C1) to (C3) define the NFW enclosed mass function, and Equation (C10) provides the generalized geometric mass ratio required for the conversion.

Notes:

The physical meaning of M_{200} and M_{500}

The choice to use values like 200 and 500 to define the mass and radius of cosmological halos (M_{200} , M_{500}) arises from a compromise between fundamental cosmological theory and the observational limits of telescopes [12, 24, 6]. In astrophysics, a cluster's mass does not have a well-defined physical "edge" [6]. Instead, the boundary is established using a density contrast (Δ) relative to the critical density of the Universe (ρ_{crit}) [6]:

$$M_{\Delta} = \frac{4}{3} \pi R_{\Delta}^3 (\Delta \cdot \rho_{\text{crit}}) \quad (79)$$

1. Why 200?

The value 200 has a theoretical origin rooted in the cosmological **spherical collapse model** [6, 18]. According to this model, a spherical density fluctuation in an Einstein-de Sitter universe (a flat universe containing only matter, $\Omega_m = 1$) collapses under its own gravity and reaches thermodynamic equilibrium (virialization) when its average density becomes exactly $18\pi^2 \approx 178$ times the background critical density [6, 18].

The step-by-step physical procedure that leads to this exact value is the following [18] (sec. 5.1.1, pag.215):

- The spherical perturbation expands until its gravity overcomes the cosmic expansion, reaching a maximum radius at the "turnaround" epoch (t_{ta}). At this exact moment, its internal density has grown to be exactly $(3\pi/4)^2 \approx 5.6$ times greater than the background density of the Universe.
- The sphere then collapses and virializes. According to the virial theorem ($2K+U=0$), the final radius of the relaxed structure is exactly half of its turnaround radius ($R_{vir} = R_{ta}/2$). Since the volume scales as the radius cubed, shrinking the radius by half increases the internal density by a factor of $2^3 = 8$.
- The collapse theoretically happens at $t_f = 2t_{ta}$. During this time, the background universe continues to expand. In a matter-dominated universe, the background density scales as t^{-2} . Therefore, from turnaround to the collapse time, the background density drops by a factor of $2^2 = 4$.
- Combining these three effects, the final density of the virialized halo compared to the background universe is exactly $\Delta_c = (3\pi/4)^2 \times 8 \times 4 = 18\pi^2 \approx 178$. [6] (sec. 2.1, eq.6, pag.81)

Astrophysicists historically rounded this value to $\Delta_c = 200$ for mathematical convenience. [6] (sec. 2.3.1, pag.85) Therefore:

- R_{200} is considered the radius within which the system is gravitationally bound and dynamically relaxed (in virial equilibrium) [6].
- M_{200} is considered the best theoretical approximation for the "total mass" of the dark matter halo [20].

2. Why 500?

While 200 is the theoretical limit, 500 is the practical limit imposed by observational reality, particularly in X-ray astronomy (e.g., using Chandra or XMM-Newton satellites) [24].

In galaxy clusters, the Intracluster Medium (ICM) emits X-rays via thermal Bremsstrahlung [24]. The luminosity of this emission scales with the square of the gas density (ρ_{gas}^2) [18]. Moving outward from the cluster center towards R_{200} , the gas density drops drastically, and its X-ray emission becomes too faint, blending completely into the cosmic background noise [24].

Telescopes can obtain reliable measurements of the gas temperature and density profiles (with a good signal-to-noise ratio) only up to approximately R_{500} (which geometrically corresponds to about 65-70% of R_{200}) [24] (sec. 7, pag.703). Furthermore, hydrodynamical simulations demonstrate that the region within R_{500} is where the gas is most confidently assumed to be in perfect **hydrostatic equilibrium**, which is a fundamental requirement for estimating mass [12] (sec.3.1,pag.9). Beyond this radius, the gas is often experiencing mildly supersonic infall, completely breaking the equilibrium assumption [12]. Therefore:

- M_{500} is the mass robustly measured through direct observations of the hot gas, without the need to extrapolate data into outer, non-hydrostatic regions [12, 24].

1.3 Virial Theorem

Simplified derivation from Mechanics

11. **Gravitational Potential Energy** (light particle m orbiting heavy particle M):

$$U = -\frac{GMm}{r} \quad (80)$$

12. **Gravitational Force:**

$$F_G = \frac{GMm}{r^2} \quad (81)$$

13. **Centripetal Force:**

$$F_c = \frac{mv^2}{r} \quad (82)$$

14. **Equating Forces** ($F_G = F_c$) to find v^2 :

$$v^2 = \frac{GM}{r} \quad (83)$$

15. **Kinetic Energy** (K , neglecting heavy particle's kinetic energy):

$$K = \frac{1}{2}mv^2 \quad (84)$$

16. **Replacing v^2 in K :**

$$K = \frac{1}{2}m \left(\frac{GM}{r} \right) \quad (85)$$

17. **Replacing Potential Energy U in K :**

$$K = -\frac{1}{2}U \quad (86)$$

18. **Virial Theorem:**

$$2 \langle K \rangle + \langle U \rangle = 0 \quad (87)$$

Cluster Virial Mass and Density

- **Context:** In a cluster, the total potential energy is the sum of all pairwise interactions, and the total kinetic energy is the sum of the kinetic energies of all galaxies. The Virial Theorem relates these totals. For an isotropic and stable system, the velocity is replaced by the velocity dispersion (σ).

19. **Total Virial Mass of the cluster** (M_{Virial}) (using σ_{3D} and characteristic radius R):

$$M_{\text{Virial}} = \frac{3 \sigma_{3D}^2 R}{G} \quad (88)$$

20. **Total Volume** (approximated as a sphere):

$$\text{Volume} = \frac{4}{3} \pi R^3 \quad (89)$$

21. **Luminous Density** (ρ_{lum}):

$$\rho_{\text{lum}} = \frac{M_{\text{lum}}}{\text{Volume}} \quad (90)$$

22. **Total Density** (ρ_{total}):

$$\rho_{\text{total}} = \frac{M_{\text{total}}}{\text{Volume}} \quad (91)$$

Data visualization

- **X-Axis:** Radius (R).
- **Y-Axis (Quantities to Plot):**
 1. ρ_{lum} (Luminous Density, Red)
 2. ρ_{total} (Total Density, Data Blue)
 3. V_{sim} (Total Simulated Velocity, Red)
 4. V_{obs} (Observed Velocity, Blue data points)

Complete Derivation

To analyze the dynamics of a self-gravitating system, such as a galaxy or a cluster of galaxies, we begin by defining its fundamental energy components. For a system of discrete interacting masses, the total gravitational potential energy U is derived from Newton's law of universal gravitation. Summing the pairwise attractive forces among all particles yields:

$$U = -G \sum_{i < j} \frac{m_i m_j}{r_{ij}} = -\frac{1}{2} G \sum_{i \neq j} \frac{m_i m_j}{r_{ij}} \quad (92)$$

where G is the gravitational constant, m_i and m_j are the masses of the individual components, and $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the distance between them. If the object is approximately spherical with a

characteristic physical radius R , the total gravitational potential energy can be parameterized as:

$$U = -\alpha \frac{GM^2}{R} \quad (93)$$

Here, $M = \sum_i m_i$ is the total mass of the system, and α is a numerical form factor of order unity that quantifies how strongly the mass is concentrated toward the center. The negative sign ensures that the gravitational interaction is strictly attractive; as the system becomes more compact ($R \rightarrow 0$), the potential energy diverges to negative infinity ($U \rightarrow -\infty$).

The total kinetic energy K of the system is simply the sum of the kinetic energies of the individual galaxies or particles:

$$K = \frac{1}{2} \sum_i m_i v_i^2 \quad (94)$$

where v_i is the velocity magnitude of the i -th component.

To connect the system's kinetic energy to its potential energy, we evaluate the scalar moment of inertia $I(t)$ of the spatial distribution of the masses [27]. The moment of inertia, which characterizes the spatial extent of the system, is defined as:

$$I(t) = \sum_i m_i \mathbf{r}_i^2(t) \quad (95)$$

where $\mathbf{r}_i$ is the position vector of the i -th mass relative to the center of mass. We now compute the time derivatives of $I(t)$. Applying the chain rule, the first derivative with respect to time is:

$$\frac{dI}{dt} = \sum_i m_i \frac{d}{dt} (\mathbf{r}_i \cdot \mathbf{r}_i) = 2 \sum_i m_i (\mathbf{r}_i \cdot \dot{\mathbf{r}}_i) \quad (96)$$

Recognizing that the time derivative of the position vector is the velocity vector ($\dot{\mathbf{r}}_i = \mathbf{v}_i$), this simplifies to:

$$\frac{dI}{dt} = 2 \sum_i m_i (\mathbf{r}_i \cdot \mathbf{v}_i) \quad (97)$$

Next, we differentiate a second time to find $\frac{d^2 I}{dt^2}$. Using the product rule on the term $(\mathbf{r}_i \cdot \mathbf{v}_i)$:

$$\frac{d^2 I}{dt^2} = 2 \sum_i m_i \frac{d}{dt} (\mathbf{r}_i \cdot \mathbf{v}_i) = 2 \sum_i m_i (\dot{\mathbf{r}}_i \cdot \mathbf{v}_i + \mathbf{r}_i \cdot \dot{\mathbf{v}}_i) \quad (98)$$

Substituting $\dot{\mathbf{r}}_i = \mathbf{v}_i$ and the acceleration $\dot{\mathbf{v}}_i = \mathbf{a}_i = \ddot{\mathbf{r}}_i$, we obtain:

$$\frac{d^2 I}{dt^2} = 2 \sum_i m_i (v_i^2 + \mathbf{r}_i \cdot \mathbf{a}_i) = 2 \sum_i m_i v_i^2 + 2 \sum_i m_i (\mathbf{r}_i \cdot \mathbf{a}_i) \quad (99)$$

According to Newton's Second Law, the net force acting on the i -th particle is $\mathbf{F}_i = m_i \mathbf{a}_i$. Substi-

tuting this into our expression yields:

$$\frac{d^2 I}{dt^2} = 2 \sum_i m_i v_i^2 + 2 \sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) \quad (100)$$

We can now express the terms of the second derivative in terms of macroscopic energy quantities. First, we identify the kinetic energy term:

$$2 \sum_i m_i v_i^2 = 4 \left(\frac{1}{2} \sum_i m_i v_i^2 \right) = 4K \quad (101)$$

Thus, substituting this back into the equation gives the general virial identity:

$$\frac{d^2 I}{dt^2} = 4K + 2 \sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) \implies \frac{1}{2} \frac{d^2 I}{dt^2} = 2K + \sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) \quad (102)$$

The second term, known as the *virial* of the system, relates to the potential energy. In a self-gravitating system, the net force $\mathbf{F}_i$ on particle i is the vector sum of all pairwise gravitational forces $\mathbf{F}_{ij}$ exerted by particles j :

$$\mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij} = \sum_{j \neq i} -G \frac{m_i m_j}{r_{ij}^3} (\mathbf{r}_i - \mathbf{r}_j) \quad (103)$$

The total virial sum is:

$$\sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) = \sum_i \sum_{j \neq i} \mathbf{r}_i \cdot \mathbf{F}_{ij} \quad (104)$$

By swapping the dummy indices i and j , and invoking Newton's Third Law ($\mathbf{F}_{ji} = -\mathbf{F}_{ij}$), we can rewrite this sum symmetrically:

$$\sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) = \frac{1}{2} \sum_i \sum_{j \neq i} (\mathbf{r}_i - \mathbf{r}_j) \cdot \mathbf{F}_{ij} \quad (105)$$

Substituting the explicit form of the Newtonian gravitational force into this symmetrical expression:

$$\sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) = \frac{1}{2} \sum_i \sum_{j \neq i} (\mathbf{r}_i - \mathbf{r}_j) \cdot \left(-G \frac{m_i m_j}{r_{ij}^3} (\mathbf{r}_i - \mathbf{r}_j) \right) \quad (106)$$

Since $(\mathbf{r}_i - \mathbf{r}_j) \cdot (\mathbf{r}_i - \mathbf{r}_j) = r_{ij}^2$, the expression collapses cleanly:

$$\sum_i (\mathbf{r}_i \cdot \mathbf{F}_i) = \frac{1}{2} \sum_i \sum_{j \neq i} -G \frac{m_i m_j}{r_{ij}} = U \quad (107)$$

Because gravity is a central force governed by a homogeneous potential of degree -1 , the virial of the forces is exactly equal to the potential energy U . Substituting this back into the second derivative equation yields the complete Virial Equation [9, 27]:

$$\frac{1}{2} \frac{d^2 I}{dt^2} = 2K + U \quad (108)$$

For a system to be stable and gravitationally bound (such as a galaxy cluster in equilibrium), its spatial scale cannot be expanding to infinity, nor can it be collapsing toward a singularity. This physical constraint means that the moment of inertia $I(t)$ is positive, finite, and bounded [10].

If we take the time-average of the Virial Equation over a sufficiently long interval, the bounded nature of $I(t)$ implies that the time-average of its second derivative must vanish [10]:

$$\left\langle \frac{d^2 I}{dt^2} \right\rangle \approx 0 \quad (109)$$

Applying this to the Virial Equation gives the formal Virial Theorem [2, 23, 5]:

$$2\langle K \rangle + \langle U \rangle = 0 \implies \langle K \rangle = -\frac{1}{2}\langle U \rangle \quad (110)$$

This indicates that for a relaxed, bound system, the kinetic energy is exactly half the magnitude of the gravitational potential energy. Additionally, the total energy of the system is $E = K + U$. Substituting the virial result yields $E = -K$, confirming that the total energy is strictly negative, an absolute requirement for the system to remain gravitationally bound.

The Virial Theorem serves as a critical tool in observational astrophysics, allowing us to estimate the dynamical mass of a system purely from observable kinematics. Returning to the balance between kinetic and potential energies (dropping the time-average brackets for steady-state systems):

$$2K = -U \implies \sum_i m_i v_i^2 = \frac{GM^2}{R} \quad (111)$$

Assuming the system consists of N objects (e.g., galaxies) of average mass m , the total mass is $M = Nm$. We can divide both sides by m to simplify the kinetic term:

$$\sum_i v_i^2 = \frac{GN^2 m}{R} \quad (112)$$

By multiplying the numerator and denominator of the right side by m , we reconstruct the total mass M :

$$\sum_i v_i^2 = \frac{GNM}{R} \implies M \sim \frac{R \sum_i v_i^2}{GN} \quad (113)$$

To make this observationally viable for a spherical cluster with an isotropic velocity distribution, we rely on the line-of-sight velocity dispersion σ_v . The 3-dimensional velocity dispersion relates to the observable 1-dimensional dispersion by $\sigma_{3D}^2 = 3\sigma_v^2$.

Replacing the summed velocity term with the statistical 3D dispersion of the cluster galaxies, the total dynamical mass, or *Virial Mass* M_{vir} , is robustly estimated as [7, 21]:

$$M_{vir} = \frac{3R\sigma_v^2}{G} \quad (114)$$

When applied to systems like the Coma Cluster, this derived dynamical mass M_{vir} is vastly greater than the sum of the luminous mass components (stars, visible gas, and dust). The dis-

crepancy provides fundamental evidence for the existence of non-luminous, gravitating Dark Matter distributed extensively beyond the visible regions [28].

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